

SYSTEM BRIEF & SECURITY SPECIFICATIONS

TrustInterview AI Platform

A Zero Trust Placement & Intelligent Interview Assessment Sandbox

1. System Architecture & Operation

TrustInterview AI operates as a split-architecture system optimized for campus placement drives. The frontend provides a responsive experience, while the backend implements security controls.

Frontend Operations (React / Vite)

- **Candidate Sandbox:** Allows students to upload resumes, parse credentials, verify CGPA eligibility, and take real-time assessments with structured prompt-driven questions.
- **Recruiter Dashboard:** A segregated command console for placement drive coordinators. Dashboards are isolated per recruiter company, displaying candidates registered or scored for that company.

Backend Operations (Node.js / Express)

- **Policy Decision Point (PDP):** Intercepts all incoming HTTP requests to evaluate authentication state, session TTL, and role-based permissions.
 - **Assessment & Grading Roster:** Orchestrates API calls to Gemini & OpenAI models. Results are compiled, graded, and stored locally in the ledger.
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2. Bias & Fairness Safeguards (ZTA Layer 9)

To ensure recruitment decisions are based entirely on technical and behavioral merits, the system employs proactive demographic and cognitive protection:

- **Dynamic Name Origin Filter:** Extracts the candidate's first and last name upon session initialization and registers them as negative filters. The AI evaluator is strictly forbidden from referencing the student's name in reviews.
 - **Demographic Redaction:** Lexical filters actively block references to Age, Gender, Race, Religion, Caste, Nationality, and Marital Status.
 - **Institutional Prestige Neutrality:** Filters terms like 'Tier-1', 'Ivy League', or 'elite college' from candidate profile inputs to ensure the grading engine remains institutionally unbiased.
 - **Bias Scanner Audits:** A lexical scanner runs on all AI outputs. If the evaluator attempts to address the candidate by name or references a protected category, the system flags a compliance violation.
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3. Anti-Hallucination & Factuality Modules (ZTA Layer 13)

The system guards against technical exaggeration and invalid claims through automated syntax audits and resume validation checks:

- **Version & Jargon Validation:** Scans student answers and resumes for non-existent tech stacks or version anomalies (e.g. Python 4.0, React v25, HTML6).
- **Metric Veracity Checks:** Flags impossible project metrics (e.g. 'reduced latency by 120%' or 'handled 10 trillion requests/sec on a Raspberry Pi').
- **Resume Mismatch Scanner:** Identifies technology stack mismatches between the interview transcript and the uploaded CV, alerting recruiters of gaps.
- **Roster Compliance Indicators:** Displays 'VERIFICATION FAILED' or " ZTA SECURED' based on the computed compliance and hallucination risk profile.

